

From FCS to DPL

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1 File change semantics

1.1 Assignments and file contexts

- An *assignment* is a total function from variables to individuals. Given a set of variables $\{x, y, z\}$, the following are all assignments:

$$\begin{bmatrix} x \rightarrow a \\ y \rightarrow b \\ z \rightarrow c \end{bmatrix} \quad \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \\ z \rightarrow a \end{bmatrix} \quad \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \\ z \rightarrow a \end{bmatrix}$$

- Given two assignments g, h , $g[x]h$ means that g and h differ only with respect to the value of x .
 - **Exercise:** given g below, list all of the assignments h , s.t., $g[y]h$. (assuming individuals a, b, c and variables x, y, z).

$$g_1 := \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \\ z \rightarrow b \end{bmatrix}$$

- A **file context** is a set of world-assignment pairs.
 - The **world set** of a file is just all of the worlds paired with some assignment in the file.
 - The **satisfaction set** of a world w in a file, is the set of assignments paired with w in the file.

1.2 Context Change Potentials

- Sentence meanings in FCS are *partial functions from files to files*, i.e., an instruction for updating some body of information.
 - Heim calls these functions ‘Context Change Potentials’.
- Since CCPs are partial functions, they may fail for certain files. Heim uses this property to account for the difference between indefinites and definites (including pronouns).

(1) A^x man walked in

$$\Rightarrow \lambda F. \begin{cases} \{ (w, g) \mid (w, g) \in F \wedge g_x \text{ a man who walked in in } w \} & \mathbf{nov}_F(x) \\ \text{undefined} & \text{else} \end{cases}$$

(2) He_x sat down

$$\Rightarrow \lambda F. \begin{cases} \{ (w, g) \mid (w, g) \in F \wedge g_x \text{ sat down in } w \} & \mathbf{fam}_F(x) \\ \text{undefined} & \text{else} \end{cases}$$

1.3 Novelty and familiarity

- The definition of **novelty** is as follows:

$$\mathbf{nov}_c(x) := \forall g, g', w [g[x]g' \rightarrow ((w, g) \in c \iff (w, g'))]$$

- If a variable isn't novel, then it is familiar.

1.4 Example 1: discourse anaphora

- A concrete example. Let's consider the flow of information in the following discourse:

(3) A^x woman walked in; she_x sat down.

- $w_\emptyset$: nobody walked in or sat down.
- w_a : a walked in and sat down; b did neither.
- w'_a : a walked and didn't sit; b didn't walk in but did sit down.
- w_b : b walked in and sat down; a did neither.
- w'_b : b walked in and sat down; a didn't walk in but did sit.
- w_{ab} : a and b both walked in and sat down.
- Variables: x, y . The initial context is the Cartesian product of logical space and the set of assignments $g : \{x, y\} \rightarrow \{a, b\}$.

$$c_0 = \left\{ \begin{array}{l} \left(w_\emptyset, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_\emptyset, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_\emptyset, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_\emptyset, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_a, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_a, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{a'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{a'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_{a'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{a'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_b, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_b, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{b'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{b'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \end{array} \right\}$$

- Update with “A^x woman walked in”.
 - Novelty is trivially satisfied.
 - We eliminate any (w, g) , where g maps x to a woman who didn’t walk in, in w .

$$c_1 = \left\{ \begin{array}{l} \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{a'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{a'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \end{array} \right\}$$

- Now let’s update c_1 with “She_x sat down”.
 - First, to check whether or not the update is defined, we check if x is familiar.
 - All that’s necessary to show that x is familiar is to find some world w which discriminates between assignments that differ only at x . There are plenty to choose from, but consider w_a , which is paired with $[x \rightarrow a, y \rightarrow a]$ but not $[x \rightarrow b, y \rightarrow a]$.
 - Now we eliminate any (w, g) , where g maps x to a woman who didn’t sit down.

$$c_2 = \left\{ \begin{array}{l} \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \end{array} \right\}$$

- We’re left just with the worlds in which someone walked in and sat down.
 - In w_a , it’s a who did so, so w_a is paired with assignments that map x to a .
 - In w_b , it’s b who did so, so w_b is paired with assignments that map x to b .

- In w_{ab} , it's both a and b who did so, so w_{ab} is paired with assignments that map a to b and assignments that map x to b .
- Note, c_2 is only true at worlds where the same woman both walked in and sat down.
- Let's double check that y still isn't familiar in c_2 (remember, we only consider assignments which differ minimally wrt y).

1.5 Example 2: concurrent indefinites

- Round two. Consider the following discourse:

(4) A^x woman walked in. A^y woman sat down.

- We already know what happens when we update the initial context with the first sentence, we get c_1 . We know that y is *novel* in c_1 , so let's update c_1 with “A^y woman sat down”.
- We eliminate any (w, g) , where g maps y to a woman who didn't sit down.

$$c_3 = \left\{ \begin{array}{c} \left(w_a, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right) \\ \left(w_{a'}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_b, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{b'}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \\ \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow a \\ y \rightarrow b \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow a \end{bmatrix} \right), \left(w_{ab}, \begin{bmatrix} x \rightarrow b \\ y \rightarrow b \end{bmatrix} \right) \end{array} \right\}$$

- c_3 is true at worlds where the same individual both walked in and sat down (w_a, w_b, w'_b, w_{ab}), but also worlds in which distinct individuals walked in and sat down (w'_a).
- FCS therefore doesn't derive *disjoint reference* out of the box, for concurrent, contra-indexed indefinites.

1.6 Interim conclusion

- FCS provides a theory of how simple utterances with indefinites and definites can be understood as *instructions for updating a body of information*.
- ‘Discourse referents’ are formalized as *variables*. The notion of a **file** formalizes the idea that participants in a discourse track the possible values of discourse referents, which are narrowed down as the discourse proceeds.
- Anaphoric links are established by assigning indefinites and definites matching variable indices.
 - Indefinites and definites effectively contribute the same information semantically, but their distribution is constrained by novelty vs. familiarity respectively.
- **Conceptual issues with FCS:**
 - Sentence meanings are functions from files to files; this is a huge departure from a classical, truth-conditional approach to meaning and blurs the boundaries between semantics and pragmatics.
 - The meanings of the logical connectives in FCS end up bearing little resemblance to their classical counterparts.

2 DPL

“The general starting point of the kind of semantics that DPL is an instance of, is that the meaning of a sentence does not lie in its truth conditions, but rather in the way it changes (the representation of) the information of the interpreter. The utterance of a sentence brings us from a certain state of information to another one. The meaning of a sentence lies in the way it brings about such a transition.” (Groenendijk & Stokhof 1991)

- Dynamic Predicate Logic (DPL) provides a dynamic interpretation schema for a simple first-order calculus.
- (Groenendijk & Stokhof 1991) give a relational presentation of DPL - formulas are interpreted relative to an evaluation point (a pair of assignments) and return a boolean value.
- Here, I give an isomorphic presentation which emphasizes two key features of the DPL interpretation schema: (i) statefulness, and (ii) non-determinism.
- Formulas are interpreted relative to an evaluation point (an assignment) and return a *set of evaluation points*.

2.1 Atomic formulas

Definition 2.1. DPL semantics for atomic formulas. For any model $M := (D, I)$

$$\llbracket t \rrbracket_M^g = \begin{cases} g_t & t \text{ is a variable} \\ I(t) & t \text{ is a constant} \end{cases}$$

$$\llbracket P(t_1, \dots, t_n) \rrbracket_M^g = \begin{cases} \{g\} & \langle \llbracket t_1 \rrbracket_M^g, \dots, \llbracket t_n \rrbracket_M^g \rangle \in I(P) \\ \emptyset & \text{otherwise} \end{cases}$$

- An example:

$$(5) \quad \llbracket \text{Swim}(x) \rrbracket^g = \begin{cases} \{g\} & g_x \text{ swims} \\ \emptyset & \text{otherwise} \end{cases}$$

- Atomic formulas in general are **tests**, i.e., identity relations on assignments.

2.2 Existential quantification

- Existential quantifiers induce what is known as **random assignment** in the dynamic literature.

Definition 2.2. DPL semantics for existential quantification.

$$\llbracket \exists x, \phi \rrbracket^g = \bigcup_{a \in D} \{g^{[x \rightarrow a]} \mid \llbracket \phi \rrbracket^{g^{[x \rightarrow a]}}\}$$

- An example, where $D = \{a, b, c\}$:

$$(6) \quad \llbracket \exists x [\text{Swim}(x)] \rrbracket^g = \{g^{[x \rightarrow a]} \mid a \text{ swims}\} \cup \{g^{[x \rightarrow b]} \mid b \text{ swims}\} \cup \{g^{[x \rightarrow c]} \mid c \text{ swims}\}$$

- Given a single evaluation point g , existential quantification gives rise to a *potentially non-deterministic output*.

Definition 2.3. Truth in DPL. A formula ϕ is *true* with respect to an assignment g iff $\llbracket \phi \rrbracket^g \neq \emptyset$

- An important fact about existential quantification in DPL: it is *externally dynamic*.

2.3 Downdate

- A curious property of DPL existential quantification is that it can destroy information established about a particular variable x .

$$(7) \quad \exists x[\text{walkedIn}(x)] \wedge \exists x[\text{satDown}(x)]$$

- Since DPL is a semantics for predicate logic, not natural language, it's unclear whether this is really a problem.
- Nevertheless, we'll attempt to get rid of this issue when we refactor FCS using ideas from DPL by incorporating a version of Heim's novelty condition.

2.4 Complex formulas

2.4.1 Conjunction in DPL

- Conjunction in DPL is interpreted as *relational composition*.

Definition 2.4. Conjunctive formula in DPL.

$$\llbracket \phi \wedge \psi \rrbracket^g = \bigcup_{h \in \llbracket \phi \rrbracket^g} \{ i \mid i \in \llbracket \psi \rrbracket^h \}$$

- This allows the indeterminate output induced by an existential quantifier to be picked up by a matching variable outside of its ordinary scope. Let's again assume that our domain consists of three individuals $\{a, b, c\}$.

$$(8) \quad \llbracket \exists x[\text{WalkedIn}(x)] \wedge \text{satDown}(x) \rrbracket^g$$

$$(9) \quad = \bigcup_{h \in \llbracket \exists x[\text{walkedIn}(x)] \rrbracket^g} \{ i \mid i \in \llbracket \text{satDown}(x) \rrbracket^h \}$$

$$(10) \quad = \bigcup_{h \in \{ g^{[x \rightarrow a]} \mid a \text{ walked in} \}} \{ i \mid i = h \wedge h_x \text{ sat down} \}$$

$$(11) \quad = \{ g^{[x \rightarrow a]} \mid a \text{ walked in and } a \text{ sat down} \}$$

- An important fact about conjunction in DPL: it is both *internally* and *externally* dynamic.
- Egli's theorem:

$$(12) \quad \exists x, \phi \wedge \psi \iff \exists x[\phi \wedge \psi]$$

2.4.2 Negation in DPL

- Negation in DPL tests whether the embedded formula is classically false.

Definition 2.5. Negated formula in DPL.

$$\llbracket \neg \phi \rrbracket^g = \begin{cases} \{g\} & \llbracket \phi \rrbracket^g = \emptyset \\ \emptyset & \text{otherwise} \end{cases}$$

- Some important facts about negation in DPL:
 - Negation is *externally static*.
 - Double-negation elimination isn't valid.
- Exercise: work through how DPL negation accounts for the facts discussed concerning the interaction between negation and anaphora.

$$(13) \quad \neg \exists x [\text{seenByJocelyn}(x)] \wedge \text{sentTheRoute}(x)$$

$$\llbracket \neg \exists x [\text{seenByJocelyn}(x)] \rrbracket^g = \begin{cases} \{g\} & \{g^{[x \rightarrow c]} \mid c \text{ was seen by Jocelyn}\} \neq \emptyset \\ \emptyset & \text{otherwise} \end{cases}$$

- But, as G&S themselves notice, the prediction for double negation doesn't seem right (see also (Krahmer & Muskens 1995); example from (Gotham 2019))

$$(14) \quad \text{John doesn't own no}^x \text{ shirt. It}_x \text{'s in his closet.}$$

- We'll come back to this problem later in the semester.

2.4.3 Material implication in DPL

- For conditional formula, we check that the outputs of the antecedent are a subset of the assignments that make the consequent classically true.

Definition 2.6. Material implication in DPL.

$$\llbracket \phi \rightarrow \psi \rrbracket^g = \begin{cases} \{g\} & \llbracket \phi \rrbracket^g \subseteq \{h \mid \llbracket \psi \rrbracket^h \neq \emptyset\} \\ \emptyset & \text{otherwise} \end{cases}$$

- An important fact about implication in DPL:
 - Implication is externally static, but internally dynamic.
- Material implication in DPL assigns existential quantifiers in the antecedent universal force. This underlies the DPL account of donkey anaphora.
- Egli's corollary: $(\exists x, \phi) \rightarrow \psi \iff \forall x[\phi \rightarrow \psi]$

(15) If a^x donkey brays, it_x is unhappy.

(16) Every donkey that brays is unhappy.

(17) $(\exists x[\text{BrayingDonkey}(x)]) \rightarrow \text{Unhappy}(x)$

- We know that the meaning of (17) is a test on g .

(18) $\llbracket \exists x[\text{BrayingDonkey}(x)] \rrbracket^g = \{ g^{[x \rightarrow a]} \mid a \text{ is a braying donkey} \}$

(19) $\llbracket \text{Unhappy}(x) \rrbracket^h = \begin{cases} \{ h \} & h_x \text{ is unhappy} \\ \emptyset & \text{otherwise} \end{cases}$

- Scenario 1: no donkeys brayed. The antecedent has an empty output and the sentence is trivially true.
- Scenario 2: Donkey d_1 brayed and is unhappy, donkey d_2 brayed and is happy.
 - The output of the antecedent contains assignments that map x to d_1 , and assignments that map x to d_2 .
 - The assignments that make the consequent classically true are just those that map x to d_1 .
 - The test fails.
- Scenario 3: Donkeys d_1 brayed and is unhappy, donkey d_2 didn't bray but is nevertheless unhappy.
 - The output of the antecedent contains assignments that map x to d_1 .
 - The assignments that make the consequent classically true are just those that map x to d_1 or x to d_2 .
 - The test passes.
- An important fact about material implication in DPL: $\phi \rightarrow \psi \iff \neg(\phi \wedge \neg \psi)$ (a classical equivalence).

- N.b. connectives can be defined under classical equivalence in DPL, but which classical equivalence matters! This is one way of understanding the explanatory challenge to dynamic semantics.
- For example, we'd get a very different (bad) result if we defined material implication in DPL as $\neg\phi \vee \psi$.
 - On the external staticity of material implication in DPL (examples from (Gillies 2019)):

(20) If a farmer owns a^x donkey, he feeds it_x hay. It_x is happy with that meal.

(21) If a farmer is happy, he feeds a^x donkey hay. ??It_x is happy with that meal.

2.4.4 Disjunction in DPL

Definition 2.7. Disjunctive formula in DPL.

$$\llbracket \phi \vee \psi \rrbracket^g = \begin{cases} \{g\} & \llbracket \phi \rrbracket^g \cup \llbracket \psi \rrbracket^g \neq \emptyset \\ \emptyset & \text{otherwise} \end{cases}$$

- An important fact about disjunction in DPL:
 - Disjunction is both *externally and internally static*.
- (22) ??Either a^x farmer left, or he_x whistled.
- (23) ??It's not the case [that a^x farmer left or that he_x whistled].
- (24) Either a^x philosopher is here, or the party is dull; ??They_x are causing a scene.
- Some well known problems for dpl disjunction: *bathroom disjunctions* (due to Barbara Partee).
- (25) Either there isn't a^x bathroom, or it_x's upstairs.
- Another problem for DPL disjunction was noted by G&S themselves. We might call them G&S disjunctions.
- (26) Either a^x philosopher was in the audience, or a^x linguist was; (Either way) they_x enjoyed it.
- Although the resolution for bathroom disjunctions is unclear, it's worth emphasizing that DPL at least has the resources to give an entry for G&S disjunctions, but it involves positing an ambiguity.

Definition 2.8. Program disjunction in DPL.

$$\llbracket \phi \cup \psi \rrbracket^g = \llbracket \phi \rrbracket^g \cup \llbracket \psi \rrbracket^g$$

2.4.5 Universal quantification

Definition 2.9. Universally quantified formula in DPL.

$$\llbracket \forall x, \phi \rrbracket^g = \begin{cases} \{g\} & \{g^{[x \rightarrow a]} \mid a \in D\} \subseteq \{h \mid \llbracket \phi \rrbracket^h \neq \emptyset\} \\ \emptyset & \text{otherwise} \end{cases}$$

A fact about universal quantification in DPL: $\forall x, \phi \iff \neg \exists \neg \phi$